

GORDON GOUGER

linkedin.com/in/gordon-gouger

gordongouger.com

EXPERIENCE

Senior Machine Learning Engineer, Ozni AI

September 2025 – Present

Machine learning lead for two SBIR programs totaling \$4M in funding.
Responsible for intelligent sensing and Pattern of Life (PoL) analytics across vision, audio, and RF modalities.
Developed and deployed drone-based geolocation algorithms leveraging onboard sensing and edge AI inference.
Designed and implemented end-to-end ML systems spanning data ingestion, model training, deployment, and field testing.
Deployed optimized models to constrained hardware including Raspberry Pi-class devices using ONNX and TensorFlow Lite.
Owned full software and hardware architecture including sensor integration, edge compute selection, and power constraints.
Built and maintained fully self-hosted CI/CD infrastructure using GitLab for model training, validation, and deployment.
Curated and managed multi-terabyte datasets for signal detection, classification, and model validation using real sensors and simulators.

Principal AI Software Engineer, Northrop Grumman

October 2021 – September 2025

Machine learning technical lead on multiple rapid prototyping and transition-to-production programs supporting operational systems.
Developed CNN-based modulation recognition models for narrowband and wideband RF, including few-shot Siamese architectures.
Deployed optimized modulation recognition pipelines combining classical DSP features (e.g., kurtosis) with CNN inference for IARPA programs.
Designed and trained transformer-based models for overhead imagery, multi-spectral, hyperspectral, and wide-band spectrum monitoring.
Built and deployed end-to-end emitter geolocation algorithms using Hawkeye 360 data, including intercept-to-link-set generation.
Performed intelligence fusion across RF, vision, and metadata sources for emitter classification and localization.
Deployed ML models to FPGA, NVIDIA Orin, and embedded edge platforms using ONNX and TensorRT.
Created and curated multi-terabyte RF datasets using real radios and high-fidelity simulators.
Built air-gapped systems utilizing locally hosted LLMs for analysis, reasoning, and operator support.
Led and mentored multiple ML engineers; served as ML lead on programs with cross-discipline teams.
Regularly interacted with customer leadership and supported deployment into live operational environments.

Engineer, Rincon Research Corporation

June 2021 – October 2021

Developed end to end GPS simulator.

EDUCATION

Master of Science in Electrical Engineering

May 2021

GPA: 4.00

Bachelor of Science in Engineering Physics

May 2020

GPA: 3.96

TECHNICAL STRENGTHS

FAA 107 Certified, Python, C++, C, MATLAB, Rust, CNNs, Transformers, Classical DSP + ML Hybrids, Modulation Recognition, Geolocation, Sensor Fusion, ONNX, TensorRT, TensorFlow Lite, Docker, Linux, GitLab CI/CD, Azure, Agile, Jira, $\LaTeX$

CLEARANCE

Active (but not currently held) TS/SCI

HOBBIES & INTERESTS

Running, Woodworking, Martial Arts, Glassblowing, Active in Local Church